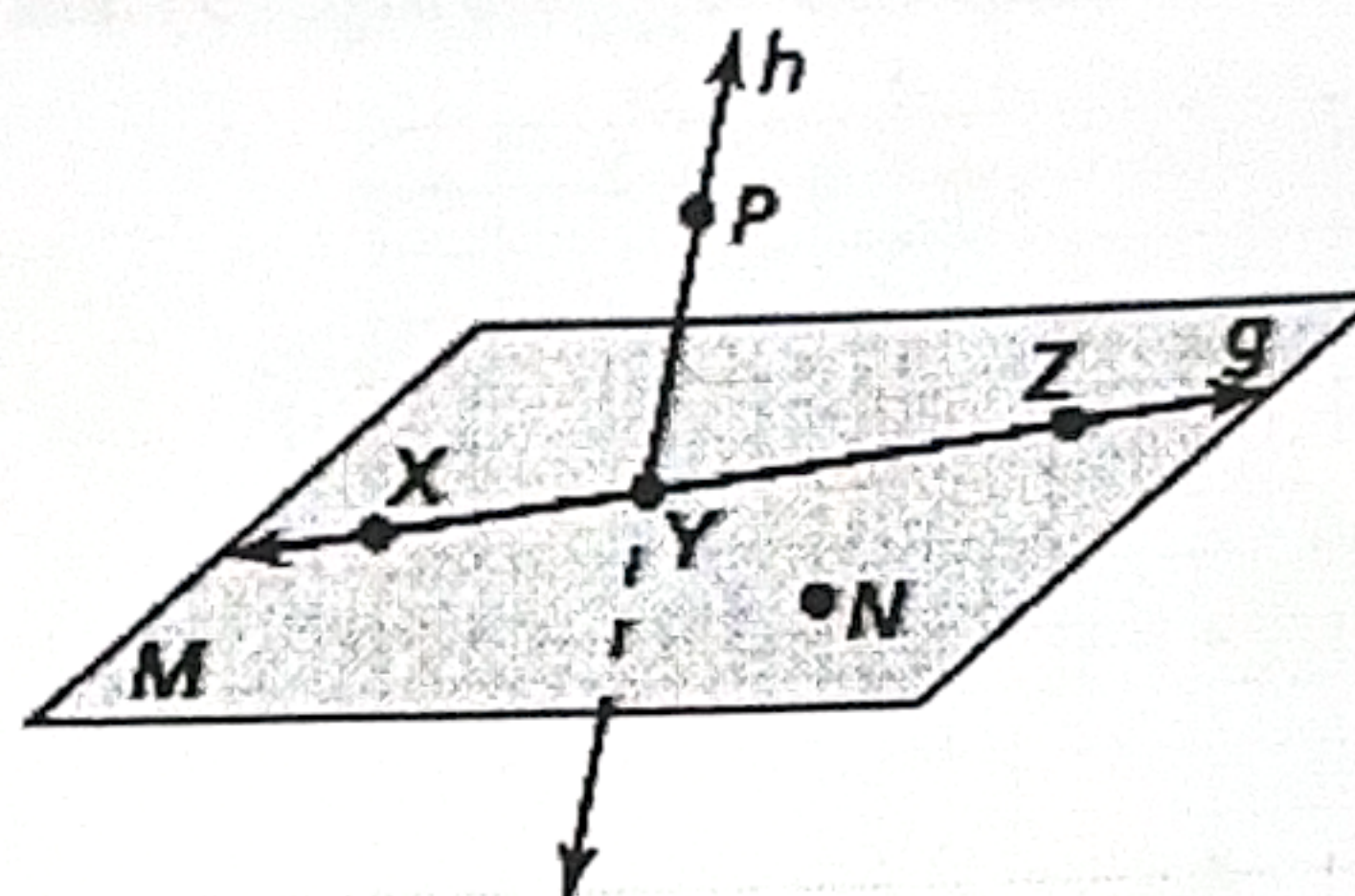


** answers may vary*
Chapter 1 Review

1.1 Understanding Points, Lines, and Planes

Use the diagram.

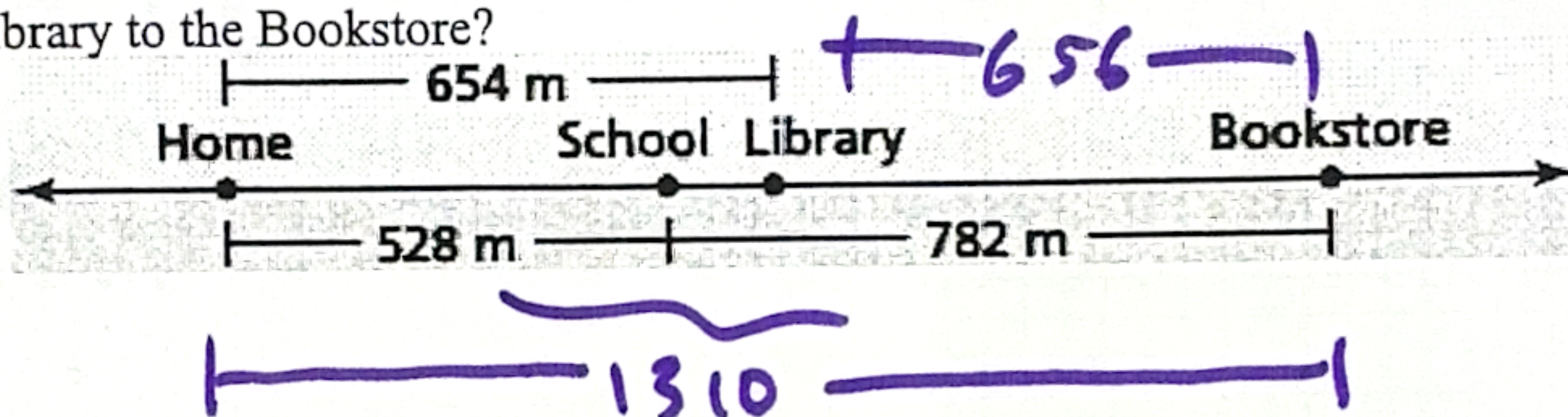
1. Give another name for Plane M. *Plane XNZ*
2. Name a line in Plane M. *$\overleftrightarrow{XZ}$*
3. Give another name for that same line. *line g*
4. Name two rays. *$\overrightarrow{YZ}$ $\overrightarrow{YX}$*



1.2 Measuring Segments

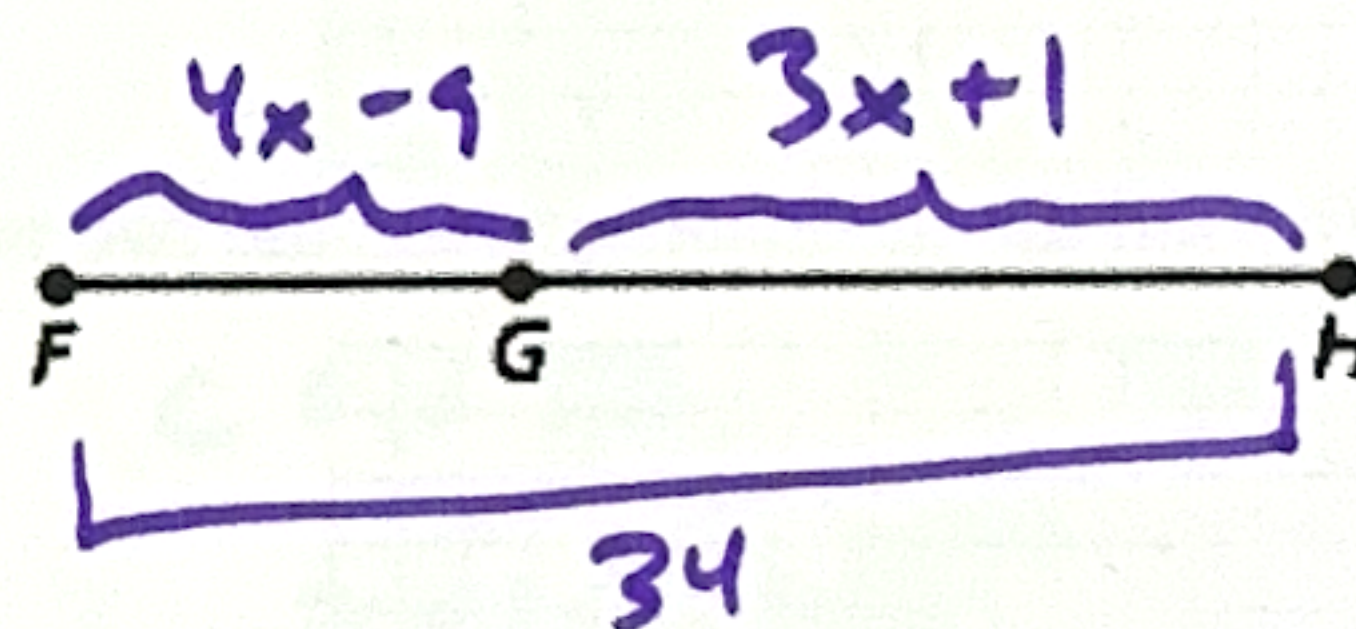
5. How many meters is it from the Library to the Bookstore?

$$1310 - 654 = \boxed{656}$$



6. Let $FG = 4x - 9$, $GH = 3x + 1$, and $FH = 34$. Find FG and GH .
a. Set up an equation that represents the relationship in the diagram.

$$4x - 9 + 3x + 1 = 34$$



- b. Solve for x .

$$\begin{array}{r} 7x - 8 = 34 \\ +8 \quad +8 \\ \hline 7x = 42 \end{array}$$

- c. Find FG and GH .

$$\frac{7x}{7} = \frac{42}{7} \quad x = 6$$

$$FG = 4(6) - 9 = \boxed{15}$$

$$GH = 3(6) + 1 = \boxed{19}$$

1.3 Using Midpoint and Distance Formulas

7. Let $AM = 3x + 8$ and $MB = 6x - 4$. Find AB .
a. Set up an equation that represents the relationship in the diagram.



- b. Solve for x .

$$\begin{array}{r} 3x + 8 = 6x - 4 \\ -3x \quad -3x \\ \hline 8 = 3x - 4 \end{array}$$

- c. Find AB .

$$\begin{array}{r} 8 = 3x - 4 \\ +4 \quad +4 \\ \hline 12 = 3x \end{array}$$

$$\frac{12}{3} = \frac{3x}{3}$$

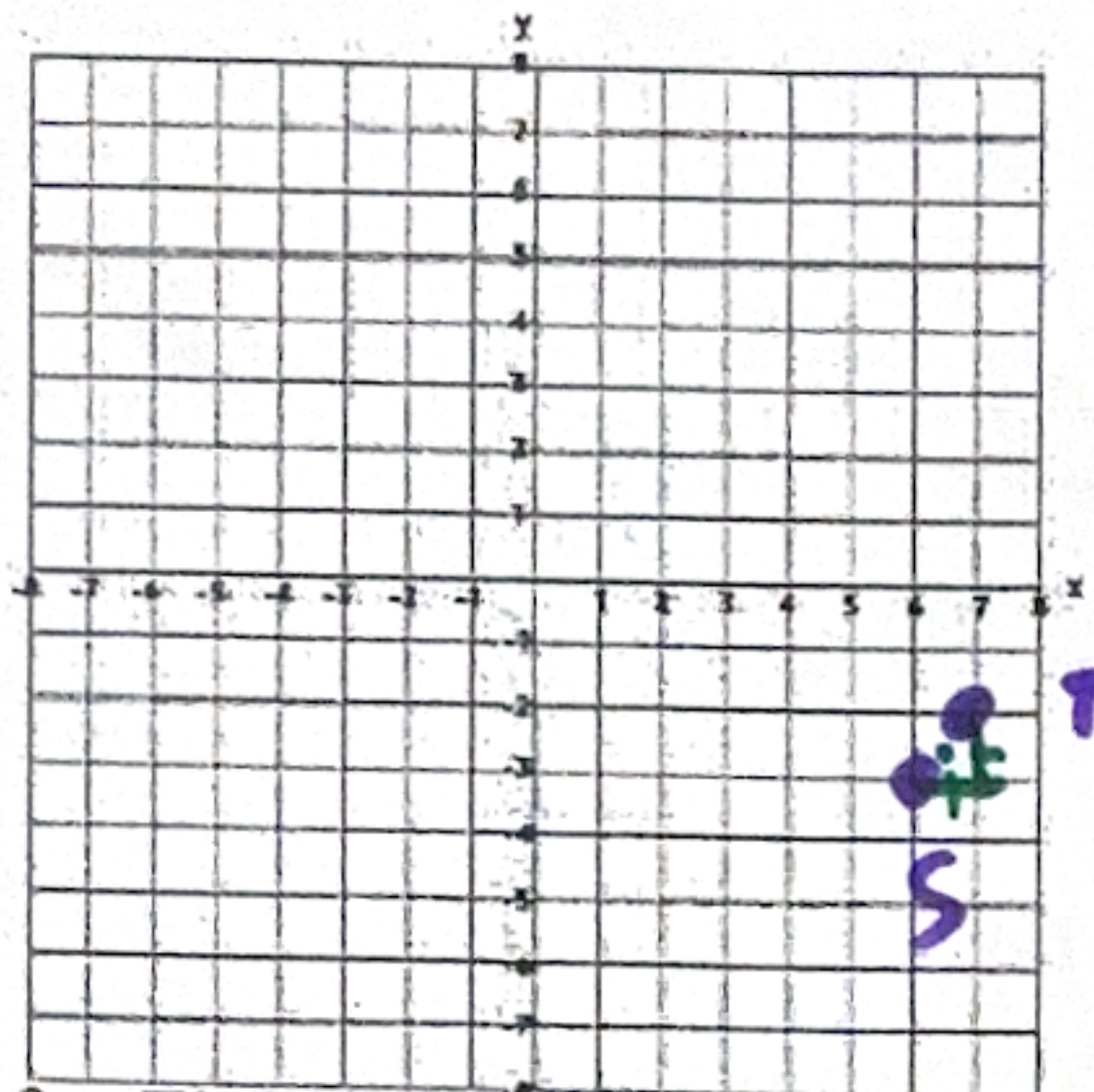
$$\boxed{x = 4}$$

$$AM = 3(4) + 8 = 20$$

$$MB = 6(4) - 4 = 20$$

$$AB = 20 + 20 = \boxed{40}$$

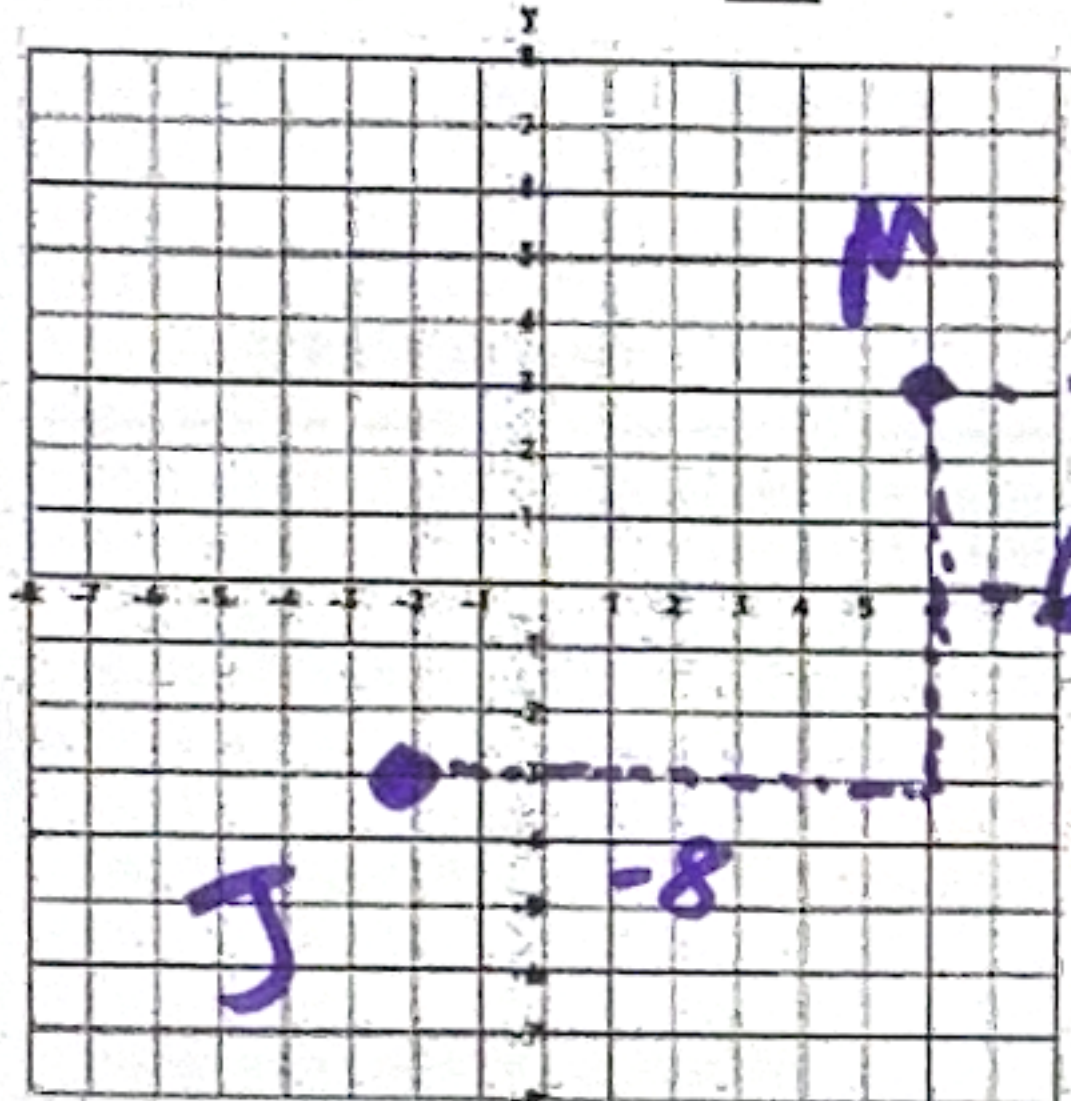
8. The endpoints of $\overline{ST}$ are $S(6, -3)$ and $T(7, -2)$. Find the coordinates for midpoint M .



$$M(6.5, -2.5)$$

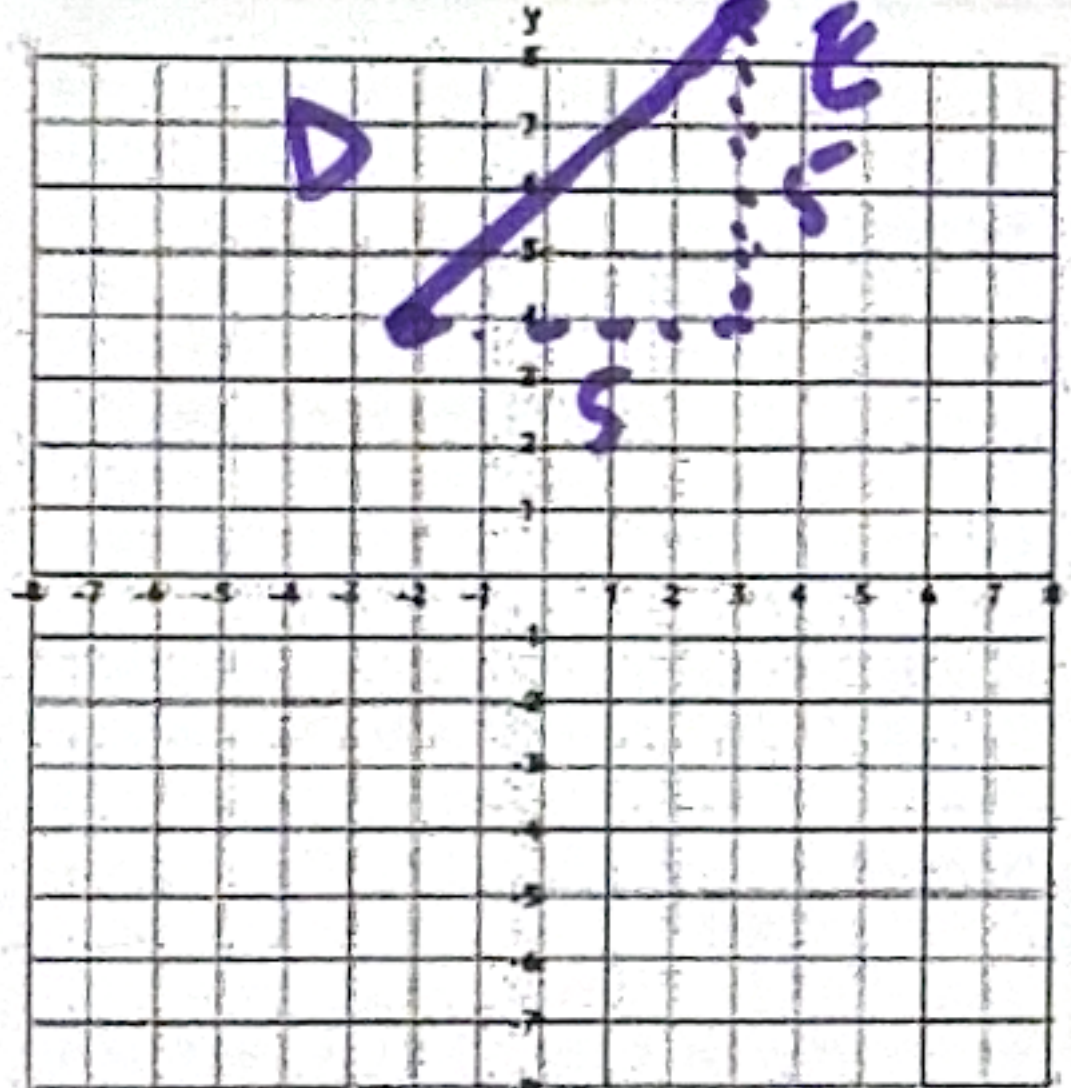
$$\begin{aligned} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{6 + 7}{2}, \frac{-3 + -2}{2} \right) \\ &= \left(\frac{13}{2}, \frac{-5}{2} \right) \end{aligned}$$

9. The midpoint of $\overline{JK}$ is $M(6, 3)$. One endpoint is $J(14, 9)$. Find the coordinates of endpoint K .



$$J(-2, -3)$$

10. Find the distance between the two points $D(-2, 4)$ and $E(3, 9)$.

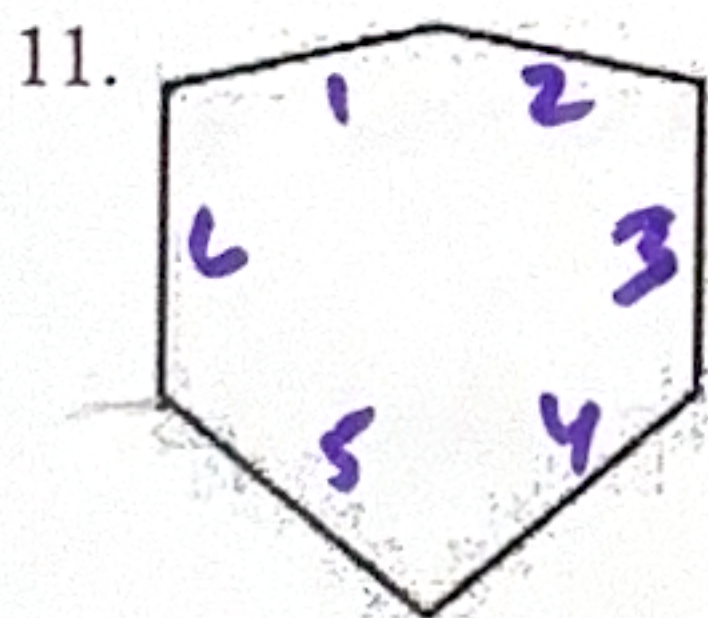


$$\begin{aligned} a^2 + b^2 &= c^2 \\ 5^2 + 5^2 &= c^2 \\ 25 + 25 &= c^2 \\ \sqrt{50} &= c \end{aligned}$$

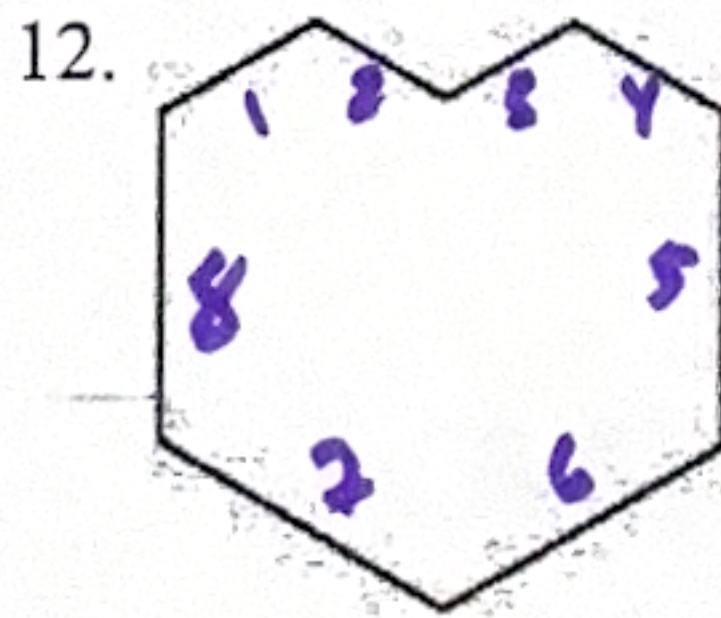
$$\begin{aligned} c &= \sqrt{50} \\ &= \sqrt{25} \sqrt{2} \\ &= 5\sqrt{2} \end{aligned}$$

1.4 Perimeter & Area in the Coordinate Plane

Classify the polygon by the number of sides.



Hexagon
(convex)



Octagon
(concave)

13. Find the perimeter and area from the polygon with vertices $W(-1, -5)$, $X(-5, -2)$, $Y(1, 6)$, and $Z(5, 3)$.

a. Find the segment lengths:

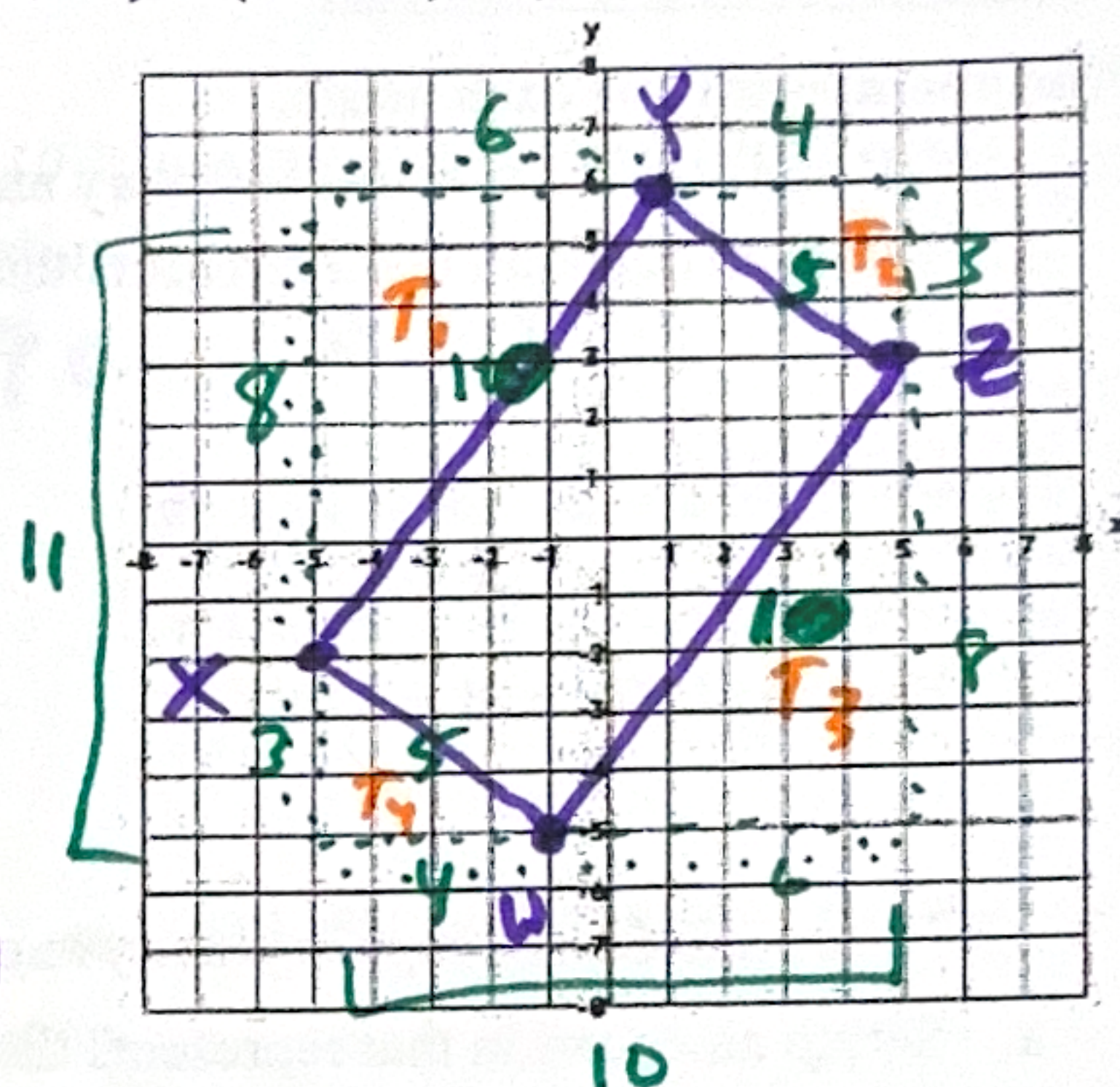
$$\begin{aligned} a^2 + b^2 &= c^2 \\ 8^2 + 6^2 &= c^2 \\ 64 + 36 &= c^2 \\ \sqrt{100} &= c^2 \\ c &= 10 \end{aligned}$$

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 3^2 + 4^2 &= c^2 \\ 9 + 16 &= c^2 \\ \sqrt{25} &= c^2 \\ c &= 5 \end{aligned}$$

$$WX = 5 \quad XY = 10 \quad YZ = 5 \quad ZW = 10$$

b. Find the Perimeter of WXYZ.

$$5 + 10 + 5 + 10 = 30 \text{ units}$$



c. Find the Area of WXYZ.

$$\begin{aligned} A_{WXYZ} &= \text{Area Rectangle} - A_{T_1} - A_{T_2} - A_{T_3} - A_{T_4} \\ &= 11 \cdot 10 - \frac{1}{2} \cdot 8 \cdot 6 - \frac{1}{2} \cdot 3 \cdot 4 - \frac{1}{2} \cdot 8 \cdot 6 - \frac{1}{2} \cdot 3 \cdot 4 = 50 \text{ units}^2 \end{aligned}$$

1.5 Measuring Angles

14. Let $m\angle ABC = 111^\circ$. Find $m\angle ABD$ and $m\angle DBC$.

a. Set up an equation that represents the relationship in the diagram.

$$-10x + 58 + 6x + 41 = 111$$

b. Solve for x .

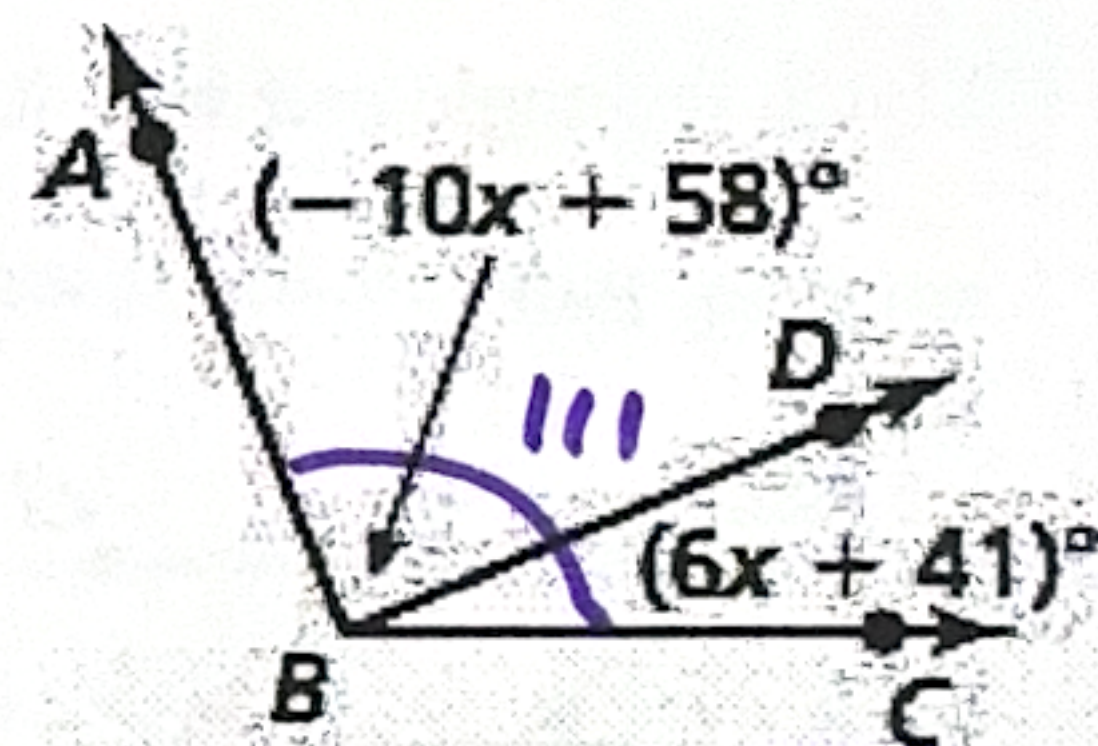
$$\begin{aligned} -4x + 99 &= 111 \\ -99 &-99 \end{aligned}$$

c. Find $m\angle ABD$ and $m\angle DBC$.

$$\begin{aligned} -4x &= 12 \\ x &= -3 \end{aligned}$$

$$m\angle ABD = -10(-3) + 58 = 88$$

$$m\angle DBC = 6(-3) + 41 = 23$$



15. Let $\overrightarrow{BD}$ bisect $\angle ABC$. Find $m\angle ABD$ and $m\angle ABC$.

a. Set up an equation that represents the relationship in the diagram.

$$\begin{aligned} 6x + 14 &= 3x + 29 \\ -3x &-3x \end{aligned}$$

b. Solve for x .

$$3x + 14 = 29$$

c. Find $m\angle ABD$ and $m\angle ABC$.

$$\begin{aligned} -14 &-14 \end{aligned}$$

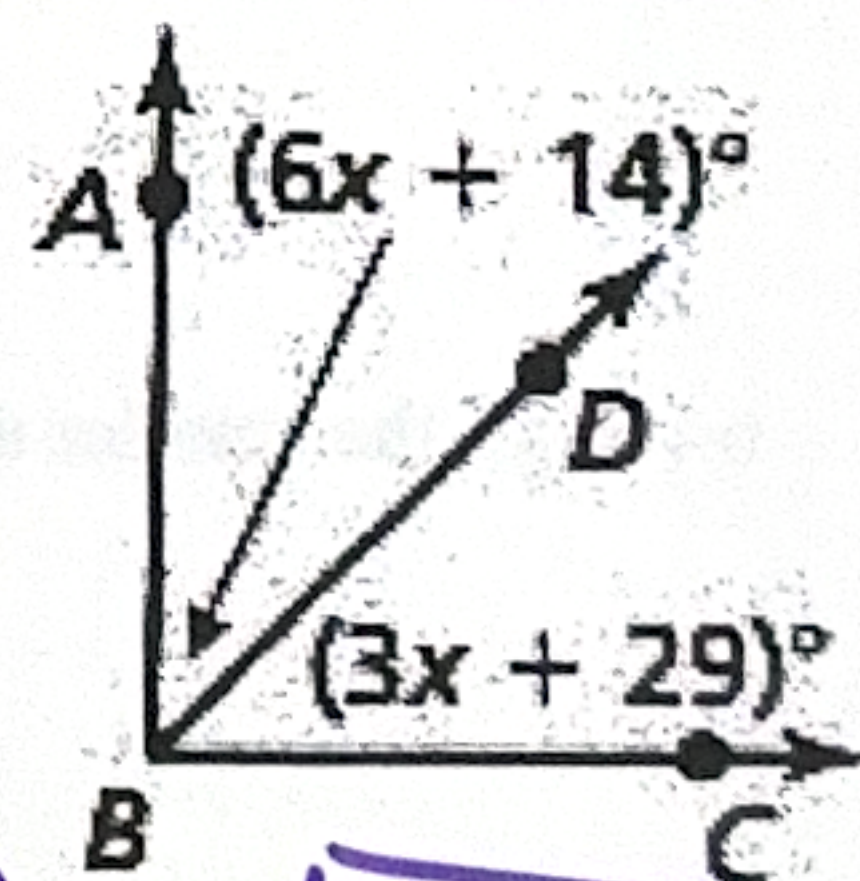
$$\begin{aligned} 3x &= 15 \\ \frac{3x}{3} &= \frac{15}{3} \end{aligned}$$

$$x = 5$$

$$m\angle ABD = 6(5) + 14 = 44$$

$$m\angle ABC = 3(5) + 29 = 44$$

$$m\angle ABC = 88$$



1.6 Describing Pairs of Angles

Find the measure of each angle.

16. $\angle UVW$ and $\angle XYZ$ are complementary angles, $m\angle UVW = (x - 10)^\circ$, and $m\angle XYZ = (4x - 10)^\circ$.

a. Set up an equation that represents the relationship in the situation.

$$x - 10 + 4x - 10 = 90 \quad x = 22$$

b. Solve for x .

$$5x - 20 = 90$$

$$m\angle UVW = 22 - 10 = 12$$

$$m\angle XYZ = 4(22) - 10 = 78$$

c. Find $m\angle UVW$ and $m\angle XYZ$.

$$\frac{5x}{5} = \frac{110}{5}$$

17. $\angle EFG$ and $\angle LMN$ are supplementary angles, $\angle EFG = (3x + 17)^\circ$, and $m\angle LMN = (\frac{1}{2}x - 5)^\circ$.

a. Set up an equation that represents the relationship in the situation.

$$3x + 17 + \frac{1}{2}x - 5 = 180$$

b. Solve for x .

$$3.5x + 12 = 180$$

$$-12 \quad -12$$

$$m\angle EFG = 3(48) + 17$$

$$= 161$$

c. Find $m\angle EFG$ and $m\angle LMN$.

$$3.5x = 168$$

$$\frac{3.5x}{3.5} = \frac{168}{3.5}$$

$$m\angle LMN = \frac{1}{2}(48) - 5$$

$$= 19$$

18. Match the angle pair with its description.

a. Linear Pair

b. Vertical Angles

c. Nonadjacent Angles

$\angle 2$ & $\angle 4$: c.

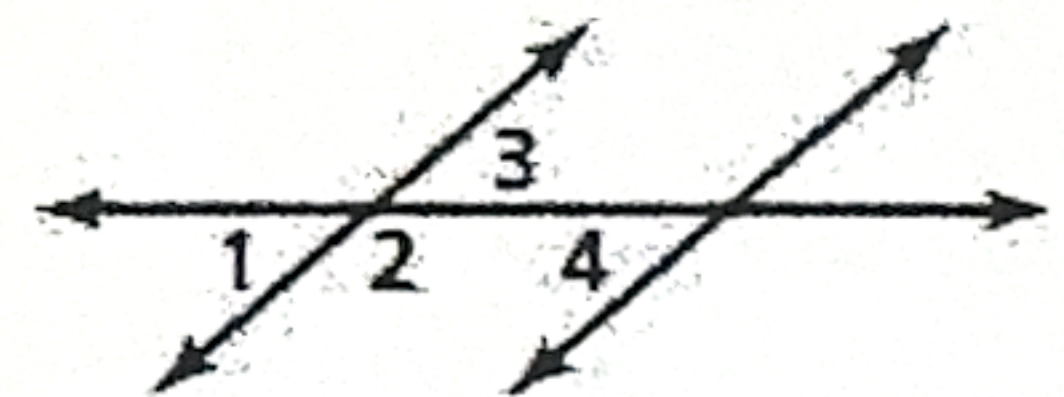
$\angle 1$ & $\angle 3$: b.

$\angle 2$ & $\angle 3$: a.

$\angle 4$ & $\angle 1$: c.

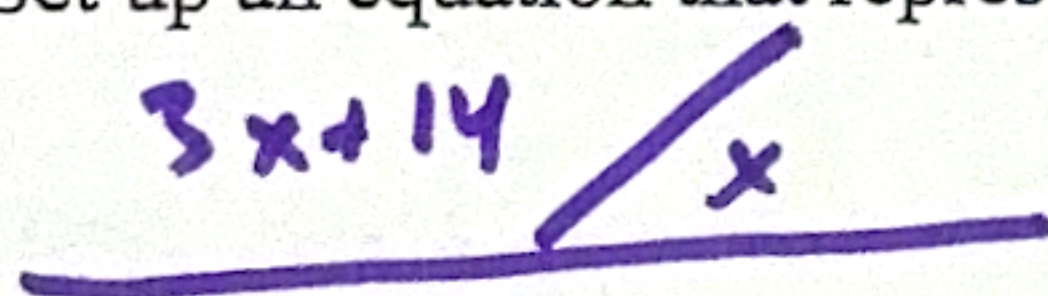
$\angle 1$ & $\angle 2$: a.

$\angle 3$ & $\angle 4$: c.



19. Two angles are complementary. The measure of the larger angle is 14° more than 3 times the measure of the smaller angle. Find the measure of both angles.

a. Set up an equation that represents the relationship described.



$$3x + 14 + x = 90$$

b. Find the smaller angle's measure.

~~measured~~

$$\text{small } \angle = 19^\circ$$

$$4x + 14 = 90$$

$$-14 \quad -14$$

$$4x = 76$$

c. Find the larger angle's measure.

$$x = 19$$

$$\text{Large } \angle = 3(19) + 14$$

$$= 71$$